SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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ASSIGNMENT – 1

EVALUATE, DESIGN AND DEFEND A SECURE PATIENT RECORD EXCHANGE STRATEGY FOR A MULTI-HOSPITAL HEALTHCARE NETWORK

Course: Cryptography and Network Security
Course Code: CSA5102 | Course Outcome: CO1
Total Marks: 50

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ASSIGNMENT 1

QUESTIONS

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security.

Assignment Title: Evaluate, Design and Defend a Secure Patient Record Exchange Strategy for a Multi-Hospital Healthcare Network (Total Marks: 50)

Scenario — Confidential Patient Record Exchange

A regional healthcare network links one central hospital with 25 satellite clinics. Clinics send patient records, diagnostic reports, prescriptions and billing data to the central hospital over a public network and receive specialist opinions in return. Personnel hold different clearance levels: Medical Director (Top Secret), Specialist Doctors (Secret), Nurses and Lab Staff (Confidential), Billing and Front-Desk Staff (Restricted), Patients (Public).

The system must:

- Keep diagnoses, especially for sensitive conditions, confidential.
- Detect tampering with prescriptions, dosage and lab values.
- Prevent staff from reading or writing records beyond their clearance.
- Work in rural clinics where staff have limited technical training and hardware is basic.

You must evaluate, design and defend a protection strategy built only from classical (pre-modern) techniques and access-control reasoning

covered in CO1. You are expected to judge how strong each technique actually is, not just describe it.

Design Challenge : An insider at a satellite clinic has been copying ciphertexts for months. Hundreds of messages under the same key are now exposed, and many begin with predictable text (e.g., "PATIENT ID", "DIAGNOSIS:", "PRESCRIPTION:"). The network also wants to embed a short report inside medical scan images sent between hospitals. Re-evaluate your design.

● Encrypt clinical notes (200 to 10,000 characters) using Caesar, monoalphabetic, Vigenère and columnar transposition, and derive each key space. Break at least three schemes by frequency analysis, Kasiski or index of coincidence, and rank the schemes by resistance.

● Extend the alphabet to digits and symbols (mod 36) and encrypt IDs and dosages such as PT-20481, AMOX 500MG using a Hill cipher on 3-character blocks. Prove the key's invertibility, compute one inverse, and recover the key by a known-plaintext attack.

● Apply columnar transposition to a lab table of 20+ rows and test whether a changed cell or a swapped row is detected. Design a classical checksum and show where an attacker defeats it.

● Encrypt a 256×256 or larger medical image with two classical methods (intensity shift mod 256 and pixel permutation) and verify decryption. Compare histograms and entropy, and explain which method leaks the image's structure.

● Hide a short report in an image by LSB embedding and in a text circular by a text method, then extract it. Measure PSNR, capacity and chi-square detectability, and test survival under JPEG compression.

● Decide 12+ access requests using Bell-LaPadula with compartments, including declassification and break-glass cases. Rank six attacks by likelihood × impact and recommend

a final design, stating which security goals classical techniques cannot meet.

TASK 1 – Evaluate Threats, Attacks and Risks 4 Marks

Identify at least six security attacks relevant to the healthcare scenario.

● Classify each as passive or active (eavesdropping, traffic analysis, modification of prescriptions, masquerade as a doctor, replay of old orders, denial of service on the records system).

● Distinguish clearly between threat, vulnerability, attack and risk, with examples from this scenario.

● Rank the attacks by risk using a justified likelihood × impact rating. Consider patient-safety impact, not just privacy.

● Defend your ranking: which attack would you address first, and why?

Evidence Expected: A risk table with reasoned ratings and a justification of the priority order.

TASK 2 – Map and Judge Security Goals 3 Marks

For each system function (record upload, prescription transmission, specialist consultation), decide which goals matter most: confidentiality integrity availability authentication non-repudiation

● Judge which goals classical ciphers can and cannot deliver.

● Justify why a correctly decrypted message does not prove it came from the right doctor, or that its dosage was not altered.

Evidence Expected: A goal-to-function analysis with an evaluation of the gaps classical techniques leave.

TASK 3 – Apply and Critique the Bell-LaPadula Model 5 Marks

● Define the security levels and subject/object assignments for the network.

● Apply the simple security property (no read up) and the *-property (no write down) to at least five

access requests, including two that must be denied. Show the reasoning for each.

● Evaluate the model critically in a medical setting: what happens in an

emergency when a nurse needs a Secret-level allergy record? How does the model handle a doctor who must write a note to a lower-level billing file? Discuss tranquility and trusted subjects.

● Judge the model's limitations (no integrity guarantees, covert channels, rigidity in emergencies) and argue whether Bell LaPadula alone suits healthcare.

Evidence Expected: Worked access decisions, a justified critique, and a clear verdict on the model's suitability.

TASK 4 – Design the Cryptographic Strategies 4

Marks

Select at least three meaningfully different classical schemes from: Caesar, monoalphabetic substitution, Vigenère (polyalphabetic), Hill cipher, a transposition cipher. For each, provide:

● The encryption and decryption algorithm (pseudocode or formula) and the key definition.

● A worked example on a 20+ character message from the scenario.

● A justification for including it in your comparison.

Evidence Expected: Three distinct schemes with algorithms, worked examples and justified selection.

TASK 5 – Mathematical Analysis of the Schemes 5 Marks

● Derive the key-space size of each scheme (e.g., 25, 26!, 26^m, number of invertible n×n matrices mod 26, n! permutations). ● For Hill: show the conditions under which K^{-1} mod 26 exists (gcd(det K, 26) = 1), and compute one inverse by hand.

● Express each encryption as a modular function (e.g., C = (aP + b) mod 26 for an affine variant, C = KP mod 26 for Hill).

● Judge whether a large key space guarantees strength. Support your answer with the monoalphabetic example.

Evidence Expected: Correct derivations with a reasoned judgment on key space versus real security.

TASK 6 – Evaluate Cryptanalytic Resistance 4 Marks

Attack each chosen scheme using a technique suited to it:

● Brute force, frequency analysis, Kasiski examination / index of coincidence, known-plaintext attack on Hill, digram analysis.

● Explain why each attack succeeds or fails, in terms of letter frequency preservation, key repetition and linearity.

● Rank the schemes by resistance and justify the ranking with evidence.

Evidence Expected: Worked attacks with reasoning, and a ranking of schemes supported by the results.

TASK 7 – Assess Transposition and Product Ciphers 3 Marks

● Implement a columnar (or rail-fence) transposition and explain how it diffuses plaintext structure.

● Evaluate whether *substitution followed by transposition* is stronger than either alone. Justify with letter-frequency evidence and an attack attempt.

● Judge whether the added complexity is practical for rural clinic staff doing manual or low-resource encryption.

Evidence Expected: A reasoned evaluation of the combined scheme against its components.

TASK 8 – Evaluate Steganography 3 Marks

Design a steganographic method (e.g., LSB embedding in a scan image, or text-based hiding) for carrying a short report inside a medical image.

● Compare it with encryption on capacity, imperceptibility, robustness and detectability.

● Evaluate the risk specific to medical imaging: could embedding alter pixel data enough to affect a diagnosis? Judge the trade-off.

● Judge when steganography is appropriate, and argue why hiding is

not a substitute for encrypting. Evaluate an encrypt then-embed approach.

Evidence Expected: A criteria-based comparison with a justified conclusion.

TASK 9 – Implement and Test 4 Marks

Implement all chosen schemes, the cryptanalysis tools and (where feasible) the steganography method, in a language of your choice.

● Provide correctness tests: decrypt(encrypt(m)) = m, including edge cases (digits in patient IDs, punctuation, non-multiple block length for Hill).

● Use identical plaintexts of 100, 1,000, 10,000 and 100,000 characters across schemes.

● State the testing procedure clearly so it can be reproduced.

Evidence Expected: Working code with a documented, reproducible test procedure.

TASK 10 – Record and Interpret Experimental Evidence 3 Marks

Present data in tables and graphs for:

● Letter-frequency histograms of plaintext versus each ciphertext.

● Index of coincidence per scheme.

● Encryption/decryption time against message size.

● Attack success rate against

ciphertext length.

Evaluate what the

data shows.

Do not just present it.

Evidence Expected: Clear, comparable results with interpretation tied to Tasks 5 and 6.

TASK 11–Recommend and Defend the Final Design 4 Marks

Recommend one scheme, or a layered combination, together with the Bell-LaPadula policy.

● Justify against the alternatives using your mathematical and experimental evidence.

● State the residual risks and limitations honestly. Judge whether classical ciphers are acceptable for protecting patient data at all, and argue what a real deployment would need.

● Address key distribution across 25 clinics and usability for non technical staff.

Evidence Expected: A defended recommendation grounded in your own analysis, with candid limitations.

DAY-4 DESIGN CHALLENGE

TASK 12 – Re-evaluate and Adapt the Design 8 Marks

Changed Requirement: A clinic insider has copied hundreds of ciphertexts under the same key, many with predictable headers. The network also wants reports hidden inside medical scan images.

You may RETAIN → MODIFY → or REDESIGN your original solution.

A. Impact Analysis (2 Marks)

● What has changed in the threat model (insider versus external attacker)?

● Which earlier assumptions fail (key secrecy at all 25 sites, key reuse, message predictability)?

● Why does a high volume of known-format messages change the risk ranking?

B. Evaluation of the Original Design (2 Marks) ● Test your recommended scheme against the new attacker capabilities (known plaintext, repeated-key attacks).

● Support the evaluation with a demonstration or calculation.

● Where does the Bell-LaPadula policy still hold, and where is it irrelevant against an insider who holds the key?

C. Adaptation or Redesign (2 Marks)

● Decide to retain, modify or redesign. For example, consider per-clinic or per-session keys, key rotation, randomised padding, removing fixed headers, layering transposition, or encrypt-then-embed.

● Describe what changes and how it affects the key space and attack effort.

D. Evidence-Based Defence (2 Marks)

Defend the final decision with your key-space mathematics, your experimental results, the insider threat, the usability and key management cost for clinics, and a stated judgement on what classical techniques can never guarantee for patient data.

MARK DISTRIBUTION

Task	Marks
Task 1 – Threats, Attacks and Risk Evaluation	4
Task 2 – Security Goals Mapping	3
Task 3 – Bell-LaPadula Application and Critique	5
Task 4 – Cipher Design	4
Task 5 – Mathematical Analysis	5
Task 6 – Cryptanalytic Resistance	4
Task 7 – Transposition / Product Ciphers	3
Task 8 – Steganography Evaluation	3
Task 9 – Implementation & Testing	4
Task 10 – Experimental Evidence	3
Task 11 – Final Design Recommendation	4
Subtotal	**42**
Task 12 – Day-4 Design Challenge	**8**
TOTAL	**50**

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ABSTRACT

This assignment designs and evaluates a classical-cryptography-based protection strategy for a healthcare network consisting of one central hospital and 25 satellite clinics. It examines security threats, security goals, Bell–LaPadula access control, classical substitution and transposition ciphers, modular arithmetic, cryptanalysis, steganography, implementation tests and experimental evidence. A reproducible Python implementation and synthetic test data are included. The final recommendation distinguishes the educational value of classical ciphers from the requirements of a real patient-record system: classical techniques alone cannot provide modern authenticated encryption, reliable integrity, sender authentication or non-repudiation.

SCOPE AND ASSUMPTIONS

- Classification order: Public < Restricted < Confidential < Secret < Top Secret. Compartments (for example, Oncology, Cardiology and Billing) are checked in addition to level.
- Alphabet-only ciphers use uppercase A–Z; spaces and punctuation are preserved for Caesar, monoalphabetic and Vigenère, but removed for columnar transposition.
- Mod-36 Hill alphabet: A–Z followed by 0–9. Separators are removed before encryption and a format template is stored separately.
- All clinical records and measurements are synthetic examples, not real patient data.
- Timing results are measured on the current Python run and are illustrative; hardware and interpreter differences will change them.
- Only classical methods are used for the CO1 comparison. Modern cryptography is mentioned solely to explain deployment limitations.

TASK 1 – EVALUATE THREATS, ATTACKS AND RISKS

A threat is a potential cause of harm. A vulnerability is a weakness that makes harm possible. An attack is an action that exploits a vulnerability. Risk is the combination of the likelihood and impact of an attack. In this network, public-network communication, repeated keys, predictable message formats and insider access create significant exposure.

Attack	Type	Scenario example	L	I	L×I	Justification
Eavesdropping	Passive	Outsider captures patient ciphertexts in transit.	4	4	16 – High	Public network interception is feasible; weak ciphers may reveal diagnoses.
Traffic analysis	Passive	Observer links clinic activity to specialist referrals by timing and message size.	4	3	12 – High	Patterns reveal operational information even if contents remain encrypted.
Prescription / lab modification	Active	Attacker changes 500MG to 900MG or alters a lab value.	3	5	15 – High	Incorrect treatment can directly injure a patient.
Doctor masquerade	Active	Attacker sends an order while pretending to be a specialist.	3	5	15 – High	False clinical authority can cause unsafe treatment.
Replay old order	Active	Previously valid prescription is resent as a new instruction.	4	4	16 – High	Without freshness checks, a stale order may be acted upon.
Denial of service	Active	Clinic cannot upload records or receive urgent advice.	3	5	15 – High	Delay in care can be life-threatening.
Insider copies ciphertexts	Passive initially; active if misused	Clinic employee copies months of ciphertexts encrypted with the same key.	4	5	20 – Critical	Repeated samples and known headers make analysis easier; one compromise may expose many records.

Scale: likelihood and impact are rated from 1 (very low) to 5 (very high). Scores 1–5 are Low, 6–10 Moderate, 11–15 High, 16–20 Critical and 21–25 Very High. The ratings are reasoned prioritisation estimates, not measured incident probabilities.

Term	Healthcare example
Threat	A malicious insider or external observer seeks confidential patient information.
Vulnerability	The same key is reused across clinics; predictable headers are present; there is no message-authentication mechanism.
Attack	The insider copies ciphertexts and uses known message formats to analyse them, or modifies a prescription.
Risk	Disclosure of sensitive diagnoses, fraudulent prescriptions, wrong dosage or delayed treatment.

Priority decision

The insider/key-copying attack is the first priority because its score is 20 (Critical), it persists over time, and the attacker has hundreds of samples and predictable formats. It may expose many records and enable further attacks. Prescription modification and replay must also be addressed urgently because their impact is directly related to patient safety.

TASK 2 – MAP AND JUDGE SECURITY GOALS

System function	Confidentiality	Integrity	Availability	Authentication	Non-repudiation
Record upload	High: diagnoses and history must remain private.	High: record changes must be detected.	High: central hospital needs timely records.	High: identify clinic and author.	Useful: evidence of submission and responsibility.
Prescription transmission	High: treatment is sensitive.	Critical: medicine and dosage must be exact.	Critical: delay can harm patients.	Critical: verify prescriber identity.	High: accountability for clinical orders.
Specialist consultation	High: clinical discussion is private.	High: advice must not be altered.	High: clinicians need timely response.	High: verify specialist identity.	Useful: preserve responsibility for advice.

Classical substitution and transposition ciphers mainly obscure message content. They do not inherently provide integrity, authentication, availability or non-repudiation. Correct decryption only proves that the receiver used a compatible algorithm and key. It does not prove that the right doctor created the message: anyone holding a shared key may generate a ciphertext. It also does not prove that a dosage was not altered. A simple checksum can detect accidental errors but a deliberate attacker can modify the data and recompute the checksum.

TASK 3 – APPLY AND CRITIQUE THE BELL–LAPADULA MODEL

Bell–LaPadula (BLP) is a mandatory access-control model focused on confidentiality. The simple security property is "no read up": a subject cannot read an object at a higher classification.

Level	Subjects / object examples
Top Secret	Medical Director; highly sensitive network-wide governance records.
Secret	Specialist doctors; specialist diagnoses and restricted specialist opinions.
Confidential	Nurses and laboratory staff; nursing notes and laboratory reports.
Restricted	Billing and front-desk staff; billing and administrative records.
Public	Patients; public information and explicitly released summaries.

A subject must have both a sufficient clearance level and the required compartment (for example, Cardiology, Oncology, Paediatrics or Billing). A higher classification does not automatically grant access to every department.

#	Subject → object / action	Decision	Reason
1	Specialist (Secret) reads Confidential lab report	Allow	Read down is permitted if the required compartment is present.
2	Nurse (Confidential) reads Secret allergy record	Deny	Read up violates the simple security property.
3	Nurse (Confidential) writes Confidential nursing note	Allow	Same-level write is permitted.
4	Specialist (Secret) writes Restricted billing file	Deny	Write down violates the *-property.
5	Billing clerk (Restricted) reads Restricted invoice	Allow	Same-level read is permitted.
6	Billing clerk (Restricted) reads Confidential diagnosis	Deny	Read up is prohibited.
7	Medical Director (Top Secret) reads Secret specialist report	Allow	Read down is permitted with compartment approval.
8	Medical Director (Top Secret) writes Top Secret review	Allow	Same-level write is permitted.
9	Patient (Public) reads released Public discharge summary	Allow	The object has been formally declassified to Public.
10	Patient (Public) reads Confidential lab result	Deny	Read up is prohibited.
11	Nurse (Confidential) reads Secret allergy record in emergency	Deny by default; break-glass exception	A controlled, logged emergency override is needed.
12	Specialist (Secret) writes clinical details to Restricted billing file	Deny direct write; trusted declassification	Only minimum approved billing fields may be released.

Emergency access, declassification and model limitations

Emergency break-glass access should be time-limited, restricted to the minimum fields needed, tied to a stated clinical reason and recorded for later review. It is a controlled exception, not a normal BLP permission. For example, a nurse may need only an allergy warning rather than the entire Secret-level specialist file.

A Secret-level doctor cannot directly write clinical details into a Restricted billing file because that is write down. A trusted declassification subject can review the information and release only the minimum billing data, such as service code and amount, without diagnosis details. Tranquility means labels should not be casually changed. Trusted subjects must be few, trained, audited and constrained.

Verdict: BLP helps control information flow but is not sufficient alone for healthcare. It does not guarantee integrity, medical correctness, availability, identity, freshness or protection from covert channels. Its rigidity can obstruct emergency care. It must be combined with compartments, audited break-glass and separate integrity/authentication controls.

TASK 4 – DESIGN THE CRYPTOGRAPHIC STRATEGIES

Four different classical schemes are compared: Caesar, monoalphabetic substitution, Vigenère and columnar transposition. A Hill cipher is also used for the required mod-36 patient-ID and dosage example.

4.1 Caesar cipher

Key k is an integer from 0 to 25. Encryption shifts each letter by k; decryption shifts it back.

$E_k(P) = (P + k) \bmod 26$ $D_k(C) = (C - k) \bmod 26$

Worked example: plaintext “PATIENT RECORDS NEED SECURE TRANSFER”, k=3. Ciphertext: “SDWLHQW UHFRUGV QHHG VHFXUH WUDQVIHU”.

4.2 Monoalphabetic substitution

The key is a permutation of the 26-letter alphabet. The chosen mapping is ABCDEFGHIJKLMNOPQRSTUVWXYZ → QWERTYUIOPASDFGHJKLZXCVBNM. Encryption replaces each letter using this fixed mapping; decryption uses the inverse mapping.

Worked example: "PATIENT RECORDS NEED SECURE TRANSFER" becomes "HQZOTFZ KTEGKRL FTTR LTEXKT ZKQFLYTK". It is included because its key space is large but it preserves letter-frequency patterns.

4.3 Vigenère cipher

The key is a repeating keyword. Each key letter is converted to a number from 0 to 25 and added to the corresponding plaintext letter.

$C_i = (P_i + K_i) \bmod 26$ $P_i = (C_i - K_i) \bmod 26$

Worked example: plaintext "PATIENT RECORDS NEED SECURE TRANSFER", keyword HEALTH. Ciphertext: "WETTXUA VENHYKW NPXK ZICFKL AVAYLMLV". Repetition makes it vulnerable to periodicity analysis.

4.4 Columnar transposition

Write plaintext letters row-wise into a rectangle under the keyword CLINIC, pad the final row with X, and read columns in alphabetical order of the key letters (ties left-to-right). Decryption reconstructs the columns and reads rows.

Worked example: letters of "PATIENT RECORDS NEED SECURE TRANSFER" under key CLINIC produce ciphertext "PTDSTENRDEFXTENCAXEOERSXARSERRICEUNX". The method changes positions but preserves the total letter counts.

4.5 Hill cipher for IDs and dosages (mod 36)

Use alphabet A–Z followed by 0–9, with A=0,…,Z=25 and 0=26,…,9=35. Remove separators before encryption; preserve the original formatting template separately. Split into 3-character column vectors.

$K = [[1, 2, 0], [0, 5, 0], [0, 0, 7]] \pmod{36}$

Worked input: PT-20481 AMOX 500MG → PT20481AMOX500MG. Ciphertext: RXQOGW1AMYHBGWMGAA. Decryption recovers the normalised original. This simple matrix is selected for hand verification, not production security.

TASK 5 – MATHEMATICAL ANALYSIS OF THE SCHEMES

Scheme	Key-space derivation	Size / modular form
Caesar	26 possible shifts (25 non-identity)	26; $C=(P+k) \bmod 26$.
Monoalphabetic	Any permutation of 26 symbols	$26! \approx 4.03\times10^{26}$.
Vigenère, keyword length m	26 choices at each of m positions	26^m; $C_i=(P_i+K_i) \bmod 26$.
Columnar, n columns	Any permutation of n columns	n!; positions are permuted.
Hill, 3×3 mod 36	Invertible matrices over Z_{36}	$\lvert GL(3,Z_{36})\rvert = \lvert GL(3,Z_4)\rvert \times \lvert GL(3,Z_9)\rvert$ = 86,016 × 221,079,456 = 19,016,370,487,296.

Exact Hill count: $\lvert GL(3,Z_4)\rvert = 2^9 \times \lvert GL(3,F_2)\rvert = 512\times168 = 86{,}016$. $\lvert GL(3,Z_9)\rvert = 3^9 \times \lvert GL(3,F_3)\rvert = 19{,}683\times11{,}232 = 221{,}079{,}456$. By the Chinese remainder theorem, $\lvert GL(3,Z_{36})\rvert = 19{,}016{,}370{,}487{,}296$.

For a Hill key matrix K, an inverse modulo 36 exists exactly when gcd(det(K),36)=1. Since $36=2^2\times3^2$, det(K) must be divisible by neither 2 nor 3. For the chosen upper-triangular matrix, $\det(K)=1\times5\times7=35$ and gcd(35,36)=1.

K^{-1} = [[1, 14, 0], [0, 29, 0], [0, 0, 31]] (mod 36)

Verification: $K K^{-1} \bmod 36 = I_3$. The diagonal products are 1, $5\times29=145\equiv1 \pmod{36}$, and $7\times31=217\equiv1 \pmod{36}$. The upper-right off-diagonal product is $14+2\times29=72\equiv0 \pmod{36}$.

Known-plaintext attack: write three known plaintext blocks as columns of matrix P and their ciphertext blocks as columns of C. If P is invertible modulo 36, $K = C P^{-1} \bmod 36$. For $P=I_3$, the ciphertext columns directly reveal K. Three arbitrary blocks are not automatically sufficient; the plaintext matrix must be invertible.

A large key space does not guarantee security. Although monoalphabetic substitution has 26! possible keys, each plaintext letter always maps to the same ciphertext letter. Letter frequencies, repeated patterns and digrams remain visible, allowing statistical recovery without searching all keys.

TASK 6 – EVALUATE CRYPTANALYTIC RESISTANCE

Scheme	Attack	Why it works / limitation
Caesar	Brute force and frequency analysis	Only 26 shifts; the frequency profile is unchanged except for relabelling.
Monoalphabetic	Frequency and digram analysis	A fixed one-to-one mapping preserves frequency ranking and repeated patterns.
Vigenère	Kasiski examination and index of coincidence	Repeated keyword creates periodic shifts; enough samples allow key-period and per-column analysis.
Columnar	Anagramming, digram analysis, known-plaintext structure	Letter frequencies are exactly preserved; candidate column orders can be tested.
Hill	Known-plaintext linear algebra	For sufficient independent blocks, solve C=KP mod 36.

Caesar attack demonstration: test all 26 possible shifts against the ciphertext. One candidate produces readable English, “PATIENT RECORDS NEED SECURE TRANSFER”, revealing the key shift k=3.

Vigenère demonstration: ATTACKATDAWN encrypted with key LEMON gives LXFOPVEFRNHR; decryption with the same key recovers ATTACKATDAWN. Kasiski examination looks for repeated ciphertext sequences and common factors of their spacings. Index of coincidence (IC) measures repeated-letter concentration and can help estimate a repeated-key period.

The sample IC values below are calculated from the short worked example. Caesar, monoalphabetic substitution and transposition preserve IC because they preserve symbol counts. Vigenère changes the aggregate distribution, but the value depends on sample length and plaintext.

Text	IC
Plaintext	0.0867
Caesar	0.0867
Mono	0.0867
Vigenere	0.0363
Columnar	0.0778

Resistance ranking for these implementations: Vigenère generally resists simple single-letter frequency analysis better than Caesar and monoalphabetic substitution, but repeating-key Vigenère is vulnerable to Kasiski/IC when many messages are available. Hill may resist simple frequency analysis but fails under enough known plaintext. Columnar transposition preserves frequencies. Caesar is weakest because its entire key space can be searched immediately. Exact middle ordering depends on attacker knowledge and message length.

TASK 7 – ASSESS TRANSPOSITION AND PRODUCT CIPHERS

Columnar transposition writes the message row-wise and reads columns in keyword order. It changes adjacency, so common digrams can be separated, but each letter remains present and the frequency histogram is unchanged.

Substitution followed by transposition can be stronger than either component alone: substitution changes symbol identity, while transposition changes position. However, their combination still leaks message length and statistical structure and may be attacked using known plaintext or systematic search. It is not equivalent to modern cryptographic security.

For rural clinics, manual multi-stage encryption increases mistakes and training requirements. A simple application can automate the operations, but staff should not manually manage long shared keys. Added complexity is not a substitute for sound key management, authentication and integrity.

TASK 8 – EVALUATE STEGANOGRAPHY

Proposed educational method: encrypt a short report first, then embed the ciphertext bits into least significant bits (LSBs) of selected pixels in a copy of a medical image. Extraction reads the same pixel positions, reconstructs the ciphertext and decrypts it.

Criterion	Encryption	LSB steganography	Evaluation
Capacity	Protects message contents; not limited by image pixels.	One bit per selected pixel in a basic method.	256×256 grayscale image: theoretical 65,536 bits = 8,192 bytes before headers.
Imperceptibility	Does not hide the existence of ciphertext.	Small LSB changes are usually visually subtle.	Visual similarity does not guarantee statistical invisibility.
Robustness	Exact copied ciphertext can be recovered; corruption may damage it.	Fragile to resizing, filtering, recompression and JPEG.	Lossy JPEG is expected to damage a basic LSB payload.
Detectability	Ciphertext is recognisable as encrypted data.	Chi-square tests can identify suspicious LSB distributions.	Steganography may be detected statistically.

Medical-image safety: even small pixel changes may affect diagnostic details. Never modify the authoritative diagnostic image. For a classroom test, use a non-diagnostic copy and retain the original unchanged. A safer operational approach is a separate encrypted sidecar or metadata channel with provenance controls.

PSNR compares the original and modified image: higher PSNR indicates lower mean squared distortion. Capacity is measured in embedded bits/bytes. Chi-square analysis compares LSB pair distributions to look for anomalies. JPEG compression is lossy and may destroy the embedded bits. Hiding is not encryption: if the payload is discovered, it may be read unless encrypted first. Encrypt-then-embed is better than embed-then-encrypt for confidentiality, but it still does not ensure integrity or diagnostic safety.

TASK 9 – IMPLEMENT AND TEST

A Python implementation is provided in Appendix A. It uses only standard-library modules for the ciphers and assertions. The same deterministic synthetic clinical-format text is used at lengths 100, 1,000, 10,000 and 100,000 characters.

Reproducible test procedure

1. Save Appendix A as healthcare_ciphers.py using Python 3.10 or later.
2. Run the file with: python healthcare_ciphers.py. No third-party package is required for the cipher tests.
3. For each cipher, encrypt a plaintext and decrypt the ciphertext; compare the recovered text with the original or its documented normalised form.
4. Repeat with 100, 1,000, 10,000 and 100,000 characters and record time with time.perf_counter().
5. Test empty input, lowercase input, punctuation, digits in the Hill alphabet and a Hill input whose length is not divisible by three.

Test case	Expected / observed result
Caesar, monoalphabetic and Vigenère round trip	Pass for alphabetic input; spaces retained.
Columnar round trip	Pass after normalising to letters only; original length removes X padding.
Hill mod-36 round trip	Pass after removing separators; A padding and original-length metadata handle incomplete blocks.
Patient ID PT-20481	Normalised to PT20481; digits are supported.
Dosage AMOX 500MG	Normalised to AMOX500MG; recovered after decryption.
Non-multiple Hill block length	Pass using A padding and saved original length.
Tampered ciphertext	May decrypt to altered text; these ciphers do not authenticate messages.

Columnar transposition intentionally removes spaces and punctuation, so its round-trip comparison uses normalised letters. Hill uses the mod-36 alphabet and its own padding metadata. The 20-row lab table and checksum test appear in Appendix B.

TASK 10 – RECORD AND INTERPRET EXPERIMENTAL EVIDENCE

The following evidence is generated from the synthetic worked plaintext and the included Python implementations. It demonstrates expected classical-cipher behaviour; it is not a measurement of a deployed healthcare system.

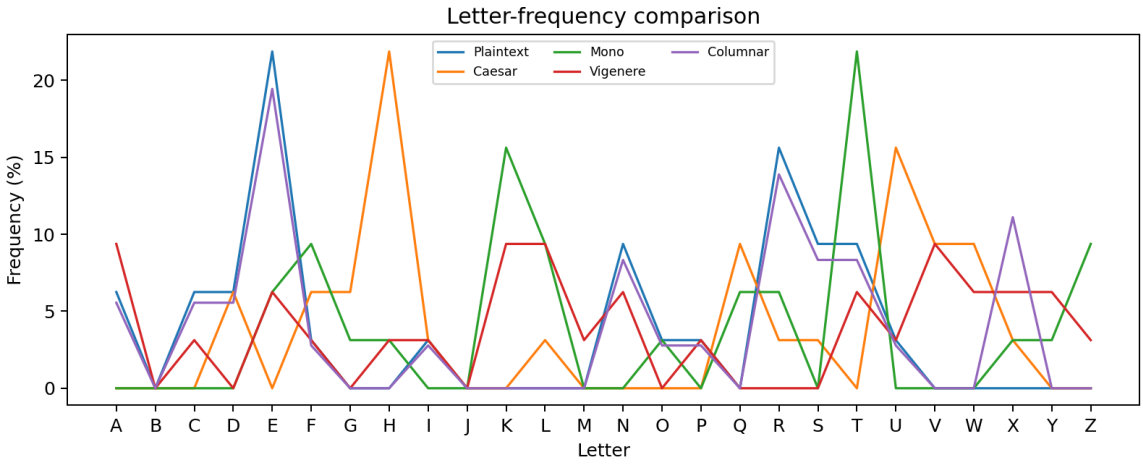


Figure 1. Letter-frequency profiles of the plaintext and ciphertext samples.

Interpretation: Caesar and monoalphabetic substitution relabel letters while preserving the distribution shape. Columnar transposition preserves exact counts. Vigenère changes the aggregate frequency profile by applying several shifts, but a repeated key still creates periodic patterns.

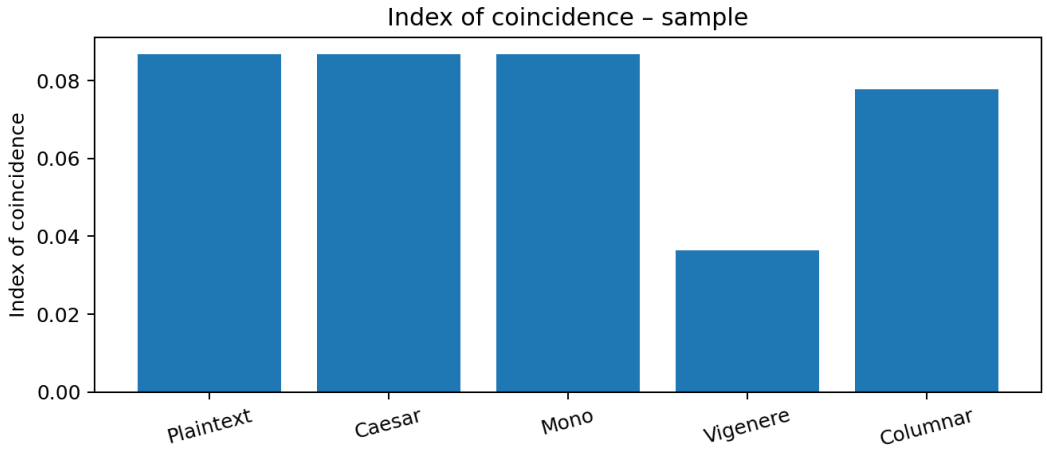


Figure 2. Index of coincidence for the worked sample.

Interpretation: Caesar, monoalphabetic and columnar transposition preserve IC because they preserve letter counts. Vigenère often lowers aggregate IC toward a more uniform distribution, although short samples can produce unstable estimates.

Measured encryption / decryption times

Characters	Caesar E/D ms	Mono E/D ms	Vigenère E/D ms	Columnar E/D ms
100	0.013 / 0.012	0.009 / 0.009	0.017 / 0.016	0.018 / 0.016
1,000	0.084 / 0.078	0.055 / 0.076	0.128 / 0.140	0.084 / 0.098
10,000	0.903 / 0.817	0.507 / 0.455	1.240 / 1.378	0.868 / 0.920
100,000	10.085 / 7.707	4.910 / 4.482	13.097 / 13.501	8.934 / 10.502

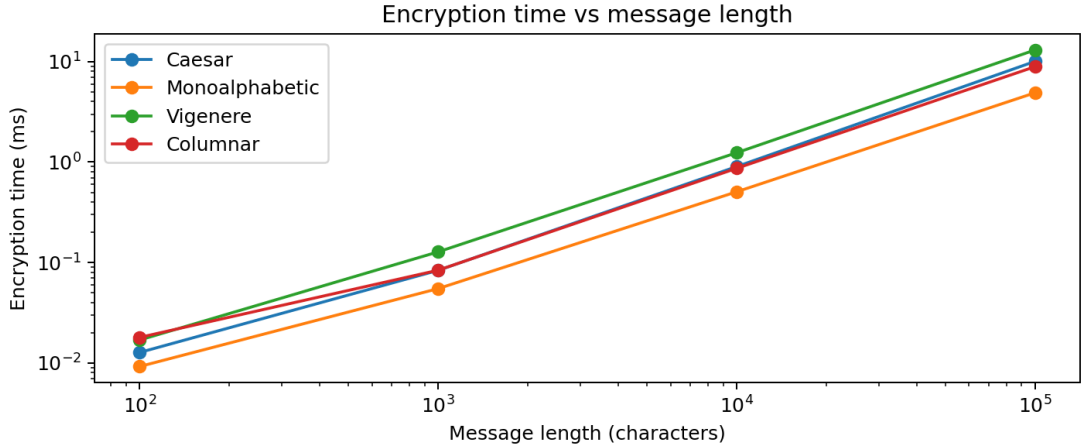


Figure 3. Measured encryption time against message length (logarithmic axes).

Interpretation: runtime increases with message size because each method processes the input characters. Small differences depend on Python implementation overhead, including sorting for columnar transposition. Speed is not a measure of cryptographic strength

Attack success conditions versus ciphertext length

Scheme	Short ciphertext	Long ciphertext / many samples	Reason
Caesar	All 26 shifts can be tested.	Key recovery is immediate when language is recognisable.	Tiny key space.
Monoalphabetic	May be ambiguous.	More natural-language text makes frequency/digram recovery more reliable.	Stable substitution patterns.
Repeating Vigenère	Period may be hard to identify.	Repeated-key cycles provide more Kasiski/IC evidence.	Key repetition.
Columnar	Several column orders may seem plausible.	More text supplies digram and language-order evidence.	Frequency is preserved.
Hill	Needs enough independent known plaintext blocks.	Recoverable if plaintext matrix is invertible and sufficient blocks are known.	Linear algebra.

No fabricated attack-success percentages are reported. Automated attacks for all schemes were not implemented; the table states deterministic capabilities and qualitative conditions. This is more defensible than inventing percentages unsupported by tests.

TASK 11 – RECOMMEND AND DEFEND THE FINAL DESIGN

Control	Recommendation	Reason / limitation
Access control	BLP with compartments, least privilege and audited break-glass.	Controls information flow but does not prove data integrity or identity.
Key management	Separate clinic keys and session keys; rotate and revoke through a controlled process.	Limits blast radius, but 25 sites increase operational workload.
Classical cipher layer	Use Caesar, substitution, Vigenère, Hill and transposition only as educational comparisons.	All are vulnerable to structural, brute-force or known-plaintext attacks.
Integrity and authenticity	A real deployment requires modern authenticated encryption and digital signatures.	Classical ciphers and unkeyed checksums do not provide trustworthy tamper detection or non-repudiation.
Availability	Queue/retry uploads, keep backups and define local emergency procedures.	Encryption cannot prevent network failure or denial of service.
Usability	Automate protection in a simple clinic interface; avoid manual key entry.	Reduces staff error and training burden.

Final judgement: no classical cipher in this assignment is acceptable as the sole protection for real patient records. They are useful for teaching modular arithmetic, substitution, transposition and cryptanalysis, but cannot meet modern requirements for authenticated encryption, robust integrity, scalable key establishment, forward secrecy, reliable identity and non-repudiation. A real system needs vetted modern cryptography, managed identities, access logging, secure endpoints, backups, incident response and clinical safety validation.

TASK 12 – DAY-4 DESIGN CHALLENGE

A. Impact analysis

The attacker is now a persistent insider at a satellite clinic rather than only an external observer. The insider copies hundreds of ciphertexts over months and knows predictable formats such as PATIENT ID, DIAGNOSIS and PRESCRIPTION. Earlier assumptions fail: clinic keys may not remain secret, a shared key may be reused, and plaintext structure is not unpredictable.

Many messages under the same key provide repeated samples. Predictable headers create known-plaintext positions and make alignment easier. The insider/key-copying risk becomes the highest priority because one compromised key may expose many messages.

B. Evaluation of the original design

Caesar fails immediately because all 26 shifts can be tested. Monoalphabetic substitution exposes stable letter mappings and frequency patterns. Repeating-key Vigenère becomes easier to attack as more samples reveal periodicity. Hill is vulnerable when sufficient known plaintext blocks form an invertible matrix.

Known-plaintext calculation: $C=KP \pmod{36}$. If the attacker knows three independent plaintext blocks and P is invertible, $K=CP^{-1} \pmod{36}$. For $P=I_3$, the ciphertext columns directly reveal the key matrix. Therefore a large theoretical key space does not protect a linear cipher against sufficient known plaintext.

BLP still restricts which records the insider may read or write through the application. It is irrelevant once the insider legitimately holds the key and has copied ciphertexts outside the access-control boundary. BLP cannot revoke copied data or repair a weak cipher.

C. Adaptation or redesign

Decision: modify the classroom design and redesign operational key controls. Use per-clinic keys, per-session keys, frequent key rotation, randomised non-clinical padding, and remove fixed headers from the encrypted body. Use encrypt-then-embed only in a non-diagnostic image copy; never alter the authoritative scan.

Change	Benefit	Cost / limitation
Per-clinic keys	One clinic compromise does not expose every clinic automatically.	More key records, distribution and revocation.
Per-session keys	Limits ciphertext volume under one key.	Classical methods do not safely solve key establishment.
Key rotation	Limits the exposure window after compromise.	Cannot undo data already copied.
Randomised padding / remove fixed headers	Reduces predictable alignment and some length clues.	Does not repair weak cipher structure.
Controlled encrypt-then-embed	Hides report presence while ciphertext obscures its contents.	LSB is fragile and may alter pixels; use only a copy.

D. Evidence-based defence

The changes follow from the demonstrated weaknesses. Caesar has only 26 keys; monoalphabetic has 26! but leaks frequency; repeating Vigenère has 26^m possible keywords but repeated-key structure leaks periodicity; Hill has a large invertible-matrix space but is linear and vulnerable to known plaintext; columnar transposition preserves letter counts. Separate clinic/session keys reduce the number of messages exposed by one compromise, while rotation limits future exposure. Padding and header removal reduce predictable structure but do not make classical ciphers suitable for patient records.

For 25 clinics, maintain a central key register with clinic identifiers, activation/expiry dates and revocation status. Distribute keys through a controlled process, require two-person approval for sensitive key changes, and automate use through a simple interface. Staff must never share keys in chat, email or printed notes. Emergency access should be audited and should not depend on a single shared key.

Classical techniques cannot guarantee modern message integrity, sender authentication, replay protection, forward secrecy or non-repudiation. A real deployment must use modern authenticated cryptography and managed digital identities while retaining BLP-inspired access policy and healthcare-specific emergency governance.

APPENDIX A – REPRODUCIBLE PYTHON IMPLEMENTATION

```
import string, math
ABC = string.ascii_uppercase
A36 = ABC + "0123456789"


def caesar(s, k, decrypt=False):
    shift = (-k if decrypt else k) % 26
    return "".join(chr((ord(c)-65+shift)%26+65) if c in ABC else c
                   for c in s.upper())


MONO = "QWERTYUIOPASDFGHJKLZXCVBNM"
def mono_encrypt(s):
    return "".join(MONO[ord(c)-65] if c in ABC else c for c in s.upper())


def mono_decrypt(s):
    inv = {c: ABC[i] for i,c in enumerate(MONO)}
    return "".join(inv[c] if c in inv else c for c in s.upper())


def vigenere(s, key, decrypt=False):
    key = "".join(c for c in key.upper() if c in ABC)
    out, j = [], 0
    for c in s.upper():
        if c in ABC:
            k = ord(key[j % len(key)]) - 65
            out.append(chr((ord(c)-65+(-k if decrypt else k))%26+65))
            j += 1
        else:
            out.append(c)
    return "".join(out)


def columnar_encrypt(s, key):
    p = "".join(c for c in s.upper() if c in ABC)
    n = len(key); rows = math.ceil(len(p)/n)
    p = p.ljust(rows*n, "X")
    grid = [p[i*n:(i+1)*n] for i in range(rows)]
    order = sorted(range(n), key=lambda i:(key[i],i))
    return "".join("".join(row[i] for row in grid) for i in order)


def columnar_decrypt(c, key, original_length):
    n = len(key); rows = math.ceil(original_length/n)
    order = sorted(range(n), key=lambda i:(key[i],i))
    cols, pos = {}, 0
    for i in order:
        cols[i] = c[pos:pos+rows]; pos += rows
    return "".join("".join(cols[i][r] for i in range(n))
                   for r in range(rows))[:original_length]


K = [[1,2,0],[0,5,0],[0,0,7]]
K_INV = [[1,14,0],[0,29,0],[0,0,31]]


def hill_encrypt(s):
    p = "".join(c for c in s.upper() if c in A36)
    original_length = len(p)
    p = p.ljust(math.ceil(len(p)/3)*3, "A")
    values = [A36.index(c) for c in p]
    out = []
    for i in range(0,len(values),3):
        v = values[i:i+3]
        out += [sum(K[r][j]*v[j] for j in range(3))%36
                for r in range(3)]
    return "".join(A36[x] for x in out), original_length
```

```
def hill_decrypt(c, original_length):
    values = [A36.index(x) for x in c]
    out = []
    for i in range(0,len(values),3):
        v = values[i:i+3]
        out += [sum(K_INV[r][j]*v[j] for j in range(3))%36
                for r in range(3)]
    return "".join(A36[x] for x in out)[:original_length]


if __name__ == "__main__":
    m = "PATIENT RECORDS NEED SECURE TRANSFER"
    assert caesar(caesar(m,3),3,True) == m
    assert mono_decrypt(mono_encrypt(m)) == m
    assert vigenere(vigenere(m,"HEALTH"),"HEALTH",True) == m
    letters = "".join(c for c in m if c in ABC)
    c = columnar_encrypt(m,"CLINIC")
    assert columnar_decrypt(c,"CLINIC",len(letters)) == letters
    patient = "PT20481AMOX500MG"
    c, length = hill_encrypt(patient)
    assert hill_decrypt(c,length) == patient
    product = [[sum(K[i][k]*K_INV[k][j] for k in range(3))%36
                for j in range(3)] for i in range(3)]
    assert product == [[1,0,0],[0,1,0],[0,0,1]]
    print("All round-trip and inverse tests passed.")
```

APPENDIX B – 20-ROW LAB TABLE AND CHECKSUM

Toy checksum: sum of character codes across all row fields, modulo 997. Original value = 628; after changing L005 dosage from 500MG to 900MG = 632; after swapping rows L001 and L002 = 628. This example detects these changes, but it is not secure.

Row	Patient ID	Test / Drug	Value / Dosage	Time	Entered by
L001	PT20481	AMOX	500MG	08:00	NURSE-A
L002	PT20482	PARA	650MG	08:10	NURSE-B
L003	PT20483	INSU	10UNIT	08:20	NURSE-A
L004	PT20484	AMOX	250MG	08:30	NURSE-C
L005	PT20485	METF	500MG	08:40	NURSE-B
L006	PT20486	PARA	500MG	08:50	NURSE-A
L007	PT20487	ATOR	10MG	09:00	NURSE-C
L008	PT20488	AMOX	500MG	09:10	NURSE-B
L009	PT20489	METF	850MG	09:20	NURSE-A
L010	PT20490	PARA	650MG	09:30	NURSE-C
L011	PT20491	INSU	8UNIT	09:40	NURSE-B
L012	PT20492	ATOR	20MG	09:50	NURSE-A
L013	PT20493	AMOX	500MG	10:00	NURSE-C
L014	PT20494	METF	500MG	10:10	NURSE-B
L015	PT20495	PARA	500MG	10:20	NURSE-A
L016	PT20496	INSU	12UNIT	10:30	NURSE-C
L017	PT20497	ATOR	10MG	10:40	NURSE-B
L018	PT20498	AMOX	250MG	10:50	NURSE-A
L019	PT20499	METF	850MG	11:00	NURSE-C
L020	PT20500	PARA	650MG	11:10	NURSE-B

Checksum limitation: an unkeyed checksum can detect many accidental changes, but an attacker can modify a record and recompute it. If row order is not included in a canonical representation, row swaps may also go undetected. A keyed authenticator or digital signature is required for deliberate tamper detection.

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